

Business Requirements Document (BRD) — Veridian Unified Data Platform

Document Reference: VCH-PC-2026-BRD | Version: 1.0 (Final Architecture Review Edition)

PROJECT CONCORD: VERIDIAN CONSOLIDATED HOLDINGS

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Target Defense Date: Friday, 14 August 2026 @ 9:00 AM Prompt

Client: Office of the Group CEO & CDTO, Veridian Consolidated Holdings

Lead Author: Israel John (Business Analyst Lead) & Samuel Enyi (Lead Database Architect)

1. Executive Summary & Project Mandate

Veridian Consolidated Holdings (VCH) is a diversified West African conglomerate operating across 5 core operating divisions: **AgriCore**, **Meridian Retail**, **Concord Logistics**, **Veridian Financial Services (VFS)**, and **Veridian Properties**. The group generates ₦340 Billion in annual revenue, employs ~14,600 staff, and serves 2.1 Million customers across Nigeria, Ghana, and Kenya.

The Business Mandate

Twenty years of organic growth and acquisitions have left VCH with 5 isolated technology stacks. This fragmentation costs VCH **4% to 6% of annual revenue (approx. ₦13.6B to ₦20.4B each year)** in stockouts, idle assets, delayed financial closing (21 calendar days), and missed cross-selling opportunities.

Project Concord mandates the design and implementation of a single, well-governed operational database—the **Veridian Core Platform**—built around a **Core Services Hub (6 shared entities)** surrounded by **5 divisional modules**, linked via scheduled ELT into **Google BigQuery** for executive analytics.

2. Business Objectives & Success Criteria

The platform must explicitly solve the **4 Critical Real-World Stakeholder Scenarios**:

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|                                     |
|               THE 4 STAKEHOLDER  |
|               SUCCESS TARGETS      |
|                                     |
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| 1. STORE MANAGER (Ngozi): Real-time alert on AgriCore harvest shortfalls to prevent |
|   stockouts at Meridian Mart supermarkets 24-48 hours before shelves go empty. |
| 2. LOAN OFFICER (Chinedu): Cross-divisional view of smallholder farmer (Ibrahim's) |
|   6-year AgriCore supply record to approve working capital loans with lower risk. |
| 3. LOGISTICS DISPATCHER (Funmi): Real-time visibility into Veridian Properties |
|   warehouse leases to prevent double-booking depot bays and shipping delays. |
| 4. GROUP CEO (Adaeze Okonji-Bello): Same-day consolidated revenue & P&L view across |
|   all 5 divisions (reducing month-end financial close from 21 days to < 24 hours).|
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3. Scope Boundaries & Constraints (Avoiding Past Failures)

To avoid the failures of VCH's 2019 attempt (monolithic scope creep, 0 tables shipped) and 2022 attempt (200+ tables, dual MySQL/Oracle, running OLAP on live OLTP):

- **In Scope:**
- **40 to 55 tables** maximum.
- PostgreSQL hosted on Supabase (OLTP System of Record).
- Google BigQuery (OLAP Analytical Warehouse).
- Row Level Security (RLS) policies and Customer Consent Flags.
- 10,000+ synthetic records covering realistic cross-divisional customer/farmer overlap.
- **Out of Scope:**
- Replacing VFS's internal core banking engine. VFS is treated as an **external read-restricted source**.
- Building mobile or web application front-ends.
- Dual-database platform implementations (MySQL/Oracle).

4. Divisional Business Rules & Schema Constraints

Below are the exact business constraints that **Samuel Enyi** must code into the SQL DDL schema (`01_schema_ddl.sql`):

4.1 Core Services Hub Rules (Shared Entities)

1. **customers** : Every individual customer must have a unique `customer_id` (e.g. `CORE_CUST_XXXXXX`), valid ISO country code (`NG` , `GH` , `KE`), and explicit consent boolean flags (`consent_retail_loyalty` , `consent_vfs_credit_sharing`).
2. **employees** : Staff members belong to exactly one division (`division_id`), with employment status `ACTIVE` , `ON_LEAVE` , or `TERMINATED` .
3. **suppliers_vendors** : Unified vendor master list. Every AgriCore farmer is registered as a supplier (`supplier_type = 'FARMER'`).
4. **locations_sites** : Universal address and geocoordinate standard (`latitude` , `longitude`) for all stores, warehouses, farms, and properties.

5. **product_service_catalogue** : Shared master SKU list (`product_id` , `unit_of_measure`) shared between AgriCore harvest outputs and Meridian Retail inventory items.
6. **financial_account_references** : Read-only reference link (`account_ref_id` , `customer_id` , `account_type` , `status`) connecting customers to VFS accounts without exposing raw banking transactions to unprivileged roles.

4.2 Divisional Module Rules

- **Meridian Retail Module:**
 - `stores` : Must reference a valid `location_id` in Core Hub.
 - `pos_transactions` : `total_amount` must be greater than zero (`CHECK (total_amount > 0)`).
 - `inventory_stock_levels` : `quantity_on_hand` cannot be negative (`CHECK (quantity_on_hand >= 0)`).
- **Concord Logistics Module:**
 - `vehicles` : Must have a valid registration number and `capacity_kg > 0` .
 - `shipments` : `destination_location_id` must differ from `origin_location_id` .
 - `warehouses` : `capacity_units` must link to an active property lease in Veridian Properties.
- **Veridian Financial Services (VFS) Module:**
 - `wallet_accounts` : `balance` cannot be negative unless an approved overdraft limit exists (`CHECK (balance >= overdraft_limit)`).
 - `loans` : `principal_amount > 0` , `interest_rate > 0` . Statuses: `APPLIED` , `DISBURSED` , `ACTIVE` , `PAID_OFF` , `DEFAULTED` (NPL = payments overdue > 90 days).
 - `kyc_records` : Tier 1, Tier 2, Tier 3 verification level strictly access-restricted under CBN microfinance guidelines.
- **AgriCore Module:**
 - `farms` : Must link to a registered farmer in Core Hub (`supplier_id`). `size_hectares > 0` .
 - `harvest_batches` : Must record `harvest_date` , `volume_kg` , and `field_agent_employee_id` .
 - `processing_runs` : Must reference a valid `harvest_id` and assign a quality grade (`GRADE_A` , `GRADE_B` , `GRADE_C` , `REJECTED`).
- **Veridian Properties Module:**
 - `leases` : `lease_end_date` must fall after `lease_start_date` (`CHECK (end_date > start_date)`).
 - `monthly_rent` must be positive.

5. Regulatory & Security Compliance Requirements

1. Central Bank Microfinance Regulation

VFS banking records are subject to strict regulatory segregation. Non-banking staff (e.g. retail store managers, logistics dispatchers) **MUST NOT** be granted direct table read access to raw VFS wallet or loan transaction tables. Access is restricted using PostgreSQL **Row Level Security (RLS)**.

2. Nigeria Data Protection Act (NDPA) & Regional Privacy Laws

Customer data sharing across divisions requires explicit consent. The database must evaluate

`consent_vfs_credit_sharing = TRUE` before exposing an AgriCore farmer's supply history to a VFS loan officer.

6. Document Sign-off & Handover

This BRD serves as the authoritative specification for **Samuel Enyi (Lead Database Architect)** to construct the `01_schema_ddl.sql` script and **Ruth Thompson** to generate synthetic CSV datasets.

- **Business Analyst Lead:** Israel John (Signed)
- **Lead Database Architect:** Samuel Enyi (Signed)
- **Data Engineering Lead:** Ruth Thompson (Signed)
- **BI Developers:** Christian Nwobodo & Richmond Akara-Nwaogu (Signed)